

Little to no active faulting likely at Europa's seafloor today

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Paul K. Byrne^{1,2} , Henry G. Dawson^{1,2}, Christian Klimczak³,
Paul V. Regensburger⁴, Kelsey T. Crane⁵, Jeffrey G. Catalano^{1,2},
Catherine M. Elder⁶, Bradford J. Foley⁷, Christopher R. German⁸,
Austin P. Green⁹, Douglas J. Hemingway¹⁰, Mohit Melwani Daswani⁶,
Mark P. Panning⁶, Noah Randolph-Flagg¹¹, Barbara Sherwood Lollar¹²,
Philip Skemer^{1,2}, Steven D. Vance⁶ & Douglas A. Wiens^{1,2}

Many of the outer Solar System's icy satellites feature known or suspected subsurface oceans, at least some of which are likely situated atop rocky interiors. Water–rock interactions at and beneath these seafloors might support active chemoautotrophic habitats, with subseafloor fluid flow facilitated by active faulting and hydrothermal systems. Absent such phenomena, however, any attainment of chemical equilibrium between the seafloor and ocean might limit the availability of chemical energy for life. Here, we characterise the stress state of the seafloor of Jupiter's moon Europa, and thus the prospect for fracturing and associated sub-seafloor fluid flow there. We consider stresses from tidal forcing, global contraction, mantle convection, and serpentinisation. We find that none of these mechanisms is likely able to drive slip along even weak, pre-existing fractures in the present. Ocean water–rock reactions taking place today are therefore probably restricted to fluid flow through only the upper few hundred metres of the seafloor. Any processes able to sustain habitable conditions at the European seafloor today must therefore be independent of ongoing tectonic activity.

On Earth, fluid-filled fracture networks can support subsurface habitable environments^{1,2}; on the ocean floors, tectonic activity drives fracturing that continually exposes fresh rock to seawater^{3–5}, allowing for the production of redox couples and the release of metabolites into the overlying water column⁶. Although the mechanical properties of the European seafloor are unknown, observations of the lithospheres of terrestrial planets imply that the brittle strength of Europa's silicate

interior is governed by the frictional properties of weaknesses inherent to large rock volumes such as pre-existing fractures at multiple scales and orientations. The magnitudes of stress will thus be limited at any given depth and pore fluid pressure by those fractures on which the resistance to frictional sliding is matched.

Here, we make a first-order characterisation of the stress state and level of geological activity of the seafloor of Europa, one of Jupiter's

¹Department of Earth, Environmental, and Planetary Sciences, Washington University in St. Louis, St. Louis, MO, USA. ²McDonnell Center for the Space Sciences, Washington University in St. Louis, St. Louis, MO, USA. ³Department of Geology, University of Georgia, Athens, GA, USA. ⁴Department of Earth Sciences, University of Oregon, Eugene, OR, USA. ⁵Seres Engineering & Services, LLC, Charleston, SC, USA. ⁶Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, USA. ⁷Department of Geosciences, Pennsylvania State University, University Park, PA, USA. ⁸Department of Geology and Geophysics, Woods Hole Oceanographic Institution, Woods Hole, MA, USA. ⁹Department of Geosciences, Virginia Polytechnic Institute and State University, Blacksburg, VA, USA. ¹⁰Institute for Geophysics, Jackson School of Geosciences, University of Texas at Austin, Austin, TX, USA. ¹¹Department of Earth and Environmental Sciences, University of Kentucky, Lexington, KY, USA. ¹²Department of Earth Sciences, University of Toronto, Toronto, ON, Canada.

 e-mail: paul.byrne@wustl.edu

largest moons and a major astrobiology target (e.g., refs. 7,8). Combining rock mechanics with spacecraft measurements, we estimate the tectonic strength of the moon's seafloor and then establish whether available stresses today are sufficient to drive failure. We further calculate the brittle lithospheric thickness of Europa's rocky interior and assess whether convection of the silicate mantle could overcome the strength of that lithosphere.

Results and discussion

Brittle strength of the rocky lithosphere

Absent any direct observations of Europa's seafloor, and in an effort to understand the *minimum* criteria required for seafloor faulting, we purposefully consider parameters that lead to the ocean floor being relatively mechanically weak—that it has been serpentinised, that it hosts pre-existing fractures, and that hydraulically connected pore spaces remain throughout the brittle portion of the moon's silicate interior.

We envisage the seafloor already being chemically altered by passive interaction with seawater⁹ or perhaps by an earlier phase of heating of the interior arising from time-varying orbital resonances with Io or changes in orbital eccentricity or tidal dissipation¹⁰. As a result, seafloor rocks are thus more likely to undergo fault slip than unaltered rock (see Table 1 and Methods). Key to this scenario is the assumption that the rocks at the seafloor are already substantially fractured from, for instance, serpentinisation and/or thermal cooling stresses (e.g., ref. 11).

By analogy with Earth, the seafloor of Europa is likely permeable at the rock–water interface (e.g., ref. 12), decreasing in permeability as depth (and overburden stress) increases. Accurately estimating a porosity profile is challenging, however¹³, and so we assume that pore space (e.g., from fracture networks inherent to large rock volumes) remains constant throughout the brittle rock volume and that these pores are hydraulically connected to the ocean above—that is, that the sub-seafloor rock is permeable and pore fluid pressure is hydrostatic. This is a conservative approach, as deviation from any of these properties or assumptions would only serve to further decrease the likelihood of faulting at Europa's ocean floor. We benchmark our modelling approach against known or proposed mechanical behaviour on Earth's seafloor and within Saturn's icy moon Enceladus (see Methods).

Our calculated limits for the brittle strength of tectonic regimes under which pre-existing and favourably oriented normal and thrust faults (i.e., pure Andersonian extension to compression, respectively) can slip are shown in Fig. 1. To demonstrate how brittle strength increases with depth, we report values for sub-seafloor depths of 100 m and 1000 m in Table 2; we illustrate the effect of pore fluid pressure on rock strength in Fig. 2. Although absolute values for lithospheric stresses at these depths are of order ~180 MPa (Fig. 1b, Table 2), we also give the tectonic stresses needed to cause faulting as differential stresses, i.e., the difference between the maximum and minimum principal compressive stresses (Table 2). At the rock–water interface itself, the stress required for failure must meet or exceed the unconfined compressive strength of the rock. For normal or thrust faulting to occur, the differential stresses at 100 m below the seafloor must reach 2.1 MPa and 2.9 MPa, respectively, increasing to 2.6 MPa and 3.5 MPa at a depth of 1000 m (Table 2).

Several mechanisms could plausibly act on pre-existing, favourably aligned fractures beneath the seafloor. For example, the 84-h diurnal tide raised by Jupiter likely drives deformation of Europa's ice shell¹⁴ and may contribute to viscous heating within the moon¹⁵. However, the maximum tidal stress at the European seafloor is currently about 54 kPa (Table 2) (see Methods), only about 3% of the lithospheric strength for extension at 100 m depth (and about the same percentage at a depth of 1000 m). The eccentricity required to drive faulting at a

Table 1 | Values for Europa model parameters in this study

Model parameter	Value
Geometric properties	
Radius (km)	1560.8
Surface gravitational acceleration (m/s ²)	1.315
Ice shell thickness (km)	30
Ocean depth (km)	97
Low-density seafloor crust thickness (km)	1
Radius of silicate mantle (km)	957.8
Radius of iron core (km)	475
Orbital properties	
Orbital eccentricity	0.009
Average orbital radius (km)	671,100
Jupiter mass (kg)	1.898×10^{27}
Mechanical properties	
Density of outer ice shell (kg/m ³)	925
Density of ocean (kg/m ³)	1025
Density of seafloor crust (kg/m ³)	2530
Density of silicate mantle (kg/m ³)	3429
Density of iron core (kg/m ³)	8000
Rock cohesion (MPa)	1.2
Rock friction coefficient	0.15
Thermal gradient (K/km)	10
Strain rate (s ⁻¹)	1×10^{-20}
Wet olivine dislocation flow law (after ref. 96)	
Material parameter, A (MPa ⁻³ s ⁻¹)	89
Stress exponent, <i>n</i>	3.5
Activation energy, <i>Q_c</i> (J/mol)	480,000
Hydrogen concentration, <i>C_{OH}</i> (H/10 ⁶ Si)	100
<i>C_{OH}</i> coefficient, <i>r</i>	1.2
Specific volume, <i>V</i> (m ³ /mol)	11×10^{-6}

depth of 1000 m beneath the seafloor is 0.441, far higher than Europa's present eccentricity of 0.009.

It is possible that sub-critical crack growth and fatigue^{16,17} resulting from repeated tidal cyclic loading over geological time has weakened the seafloor rocks. Under the assumption that cycling reduces short-term, normal fault strength by about 75% (ref. 18), the modern diurnal tidal stress may come within a factor of ten of matching the strength of fatigued rocks at 100 m beneath the seafloor, but even then is twelve times too low to induce failure at a subseafloor depth of 1000 m.

Global contraction from secular interior cooling is accommodated in the brittle silicate portion of a body by the formation of thrust faults¹⁹. We find that the European silicate interior would need to contract radially by 1 km before the elastic strength of the seafloor is matched²⁰ and thrust faulting can begin (see Methods). This value is greater than that found for the Moon (~several tens of metres²¹) but less than estimates for Mars (≤ 3.8 km, ref. 22) or Mercury (≤ 7 km, ref. 23). The timing of such contraction of Europa's silicate interior, and when and for how long faults might have been actively accommodating contractional strain, remains to be quantified. Yet the record of global contraction on Mercury, for example, points to substantial radius decrease within the first billion years of the planet's history²⁴, and a reduction in silicate radius of 1 km over Solar System history gives a strain rate of only 5×10^{-21} s⁻¹. Moreover, the sealing of faults by gouge and, in submarine settings especially, by rapid mineral precipitation will serve to substantially decrease fault permeability in the absence of sustained tectonic activity²⁵. Therefore, even if thrusts developed in the past from secular cooling, it is far from clear whether

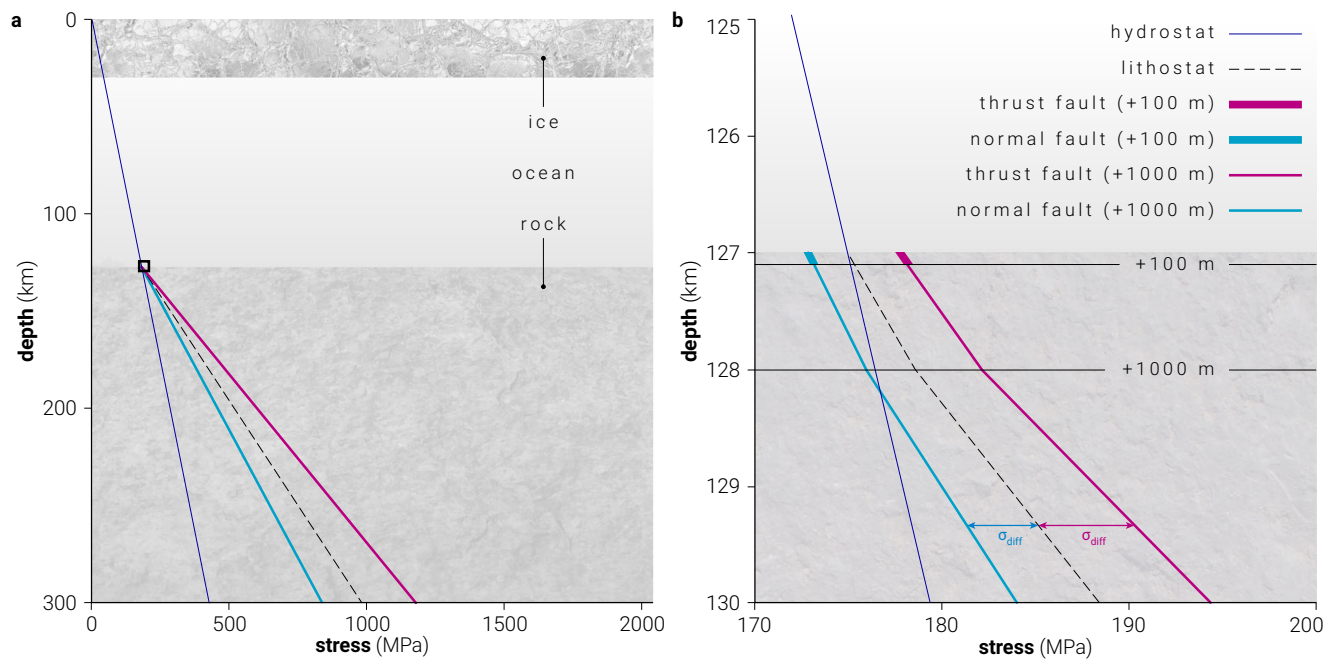


Fig. 1 | Brittle strength profiles. **a** Brittle strength for Europa’s rocky lithosphere under extensional and contractional tectonic regimes as a function of depth (blue and pink lines, respectively), together with hydrostatic and lithostatic stress (thin navy and dashed black lines, respectively). The small black box indicates the area shown in detail in panel **(b)**. **b** An expanded view of the inset in panel **a**, showing

brittle strength as a function of depth for the upper 3 km of Europa’s silicate interior; depths of 100 m and 1000 m are marked. Absolute stress values are given along the x-axes. The differential stresses (σ_{diff}) required for driving normal (S_V-S_H) and thrust (S_H-S_V) faulting are illustrated in **b**.

Table 2 | Europa model results in this study

Model result	Value
Lithostatic strength (100 m) (MPa)	175
Lithostatic strength (1000 m) (MPa)	179
Effective stress (100 m) (MPa)	0.2
Effective stress (1000 m) (MPa)	2.2
Lithospheric strength in a normal fault regime (100 m) (MPa)	173
Lithospheric strength in a normal fault regime (1000 m) (MPa)	176
Differential lithospheric strength in a normal fault regime (100 m) (MPa)	2.1
Differential lithospheric strength in a normal fault regime (1000 m) (MPa)	2.6
Lithospheric strength in a thrust fault regime (100 m) (MPa)	178
Lithospheric strength in a thrust fault regime (1000 m) (MPa)	182
Differential lithospheric strength in a thrust fault regime (100 m) (MPa)	2.9
Differential lithospheric strength in a thrust fault regime (1000 m) (MPa)	3.5
Maximum tidal stress (kPa)	54
Change in silicate radius to match thrust fault strength (km)	−1.0
Brittle–plastic transition depth in a normal fault regime (km)	56
Brittle–plastic transition depth in a thrust fault regime (km)	55
Peak lithospheric strength for normal faults (at depth below seafloor) (MPa)	51 (56 km)
Peak lithospheric strength for thrust faults (at depth below seafloor) (MPa)	67 (55 km)
Mantle stress (10^{18} Pa s) (kPa)	35
Mantle stress (10^{20} Pa s) (kPa)	150
Thermal expansion anisotropy (MPa)	3

such faults might be active today or be able to conduct seawater to unaltered rock.

Fluid and heat transport persist in very low-permeability regions of Earth’s seafloor, facilitated by microfracture channels in the rock matrix that form by thermal expansion anisotropy between mineral grains. In the absence of widespread slip on pre-existing fractures, microfracturing may be the primary enabler of aqueous diagenesis in water-rich bodies such as Ceres and Europa^{26,27}. Microfractures develop along grain boundaries because of stress concentrations from flaws such as precipitates, and form when the stress intensity at the tip of the grain exceeds some critical value (see Methods). For a conservatively low cooling rate of 1 K/Gyr, the normal stress on a two-dimensional grain boundary²⁸ can exceed 3 MPa. Assuming that microfracturing develops parallel to the intermediate principal compressive stress, this process serves to add to the local differential stress and so might help facilitate slip—but only at shallow crustal levels. We calculate that increasing the differential stress by 3 MPa may contribute to faulting only to a subseafloor depth of 2.5 km and 3 km for normal and thrust faults, respectively. (These values are for rocks with hydrostatic pore fluid pressure; without it, thermal cracking has basically no effect upon rock strength.)

On Earth, observations of triggered and induced seismicity suggest that many faults have elevated pore pressures, are critically stressed, and are sensitive to kPa-scale changes from hydrogeological loads, tides, and teleseismic waves (e.g., refs. 29,30). Such structures, aided by locally higher-porosity damage zones, represent pathways for fluid transport even in the absence of active slip. This flow is generally driven by large thermal gradients³¹, compaction of saturated materials, or dehydration of minerals³². Although the silicate interior could still be dehydrating³³, Europa’s relatively low gravity has been proposed to substantially reduce the vigour of gravity-driven fluid advection³⁴, and the relatively high overburden suggests a reduction of permeability with depth of about two orders of magnitude relative to smaller bodies

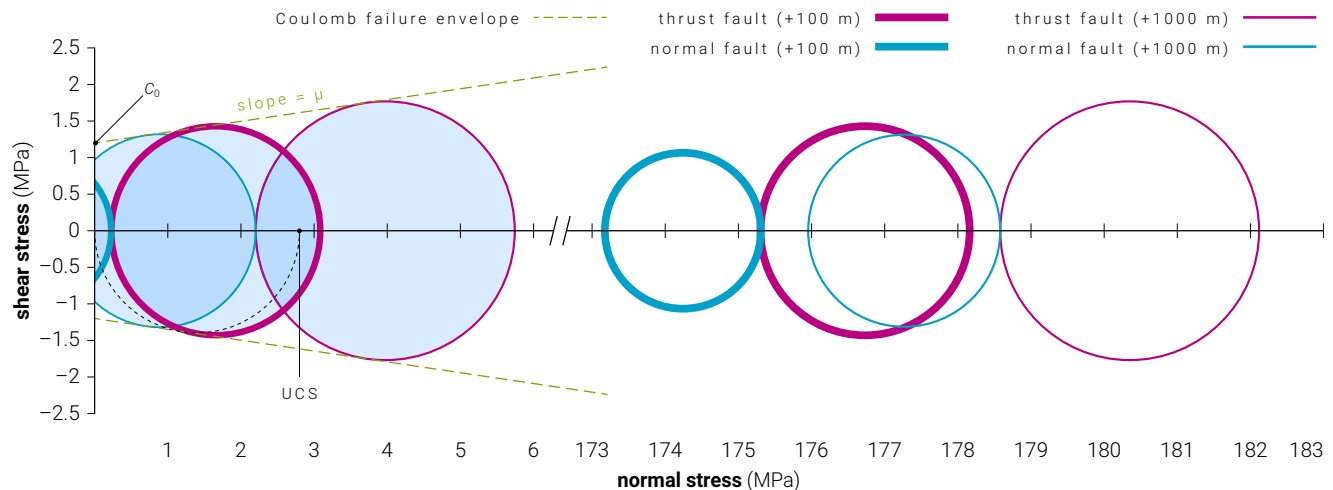


Fig. 2 | The effect of pore fluid pressure on rock strength. Mohr circles illustrating the stress state for normal (blue) and thrust (magenta) faults at a sub-seafloor depth of 100 m (thick circles) and 1000 m (thin circles). Rock strength is substantial in the absence of pore fluid (unfilled circles, right), but drops considerably when counteracted by hydrostatic pore fluid pressure (filled circles, left).

In this diagram, the Coulomb failure envelope is shown by a dashed green line; C_0 marks rock cohesion (1.2 MPa), the slope of the failure envelope is the rock coefficient of internal friction (μ , 0.15), and UCS is the unconfined compressive strength of the rock (~2.8 MPa).

such as Enceladus³⁵. The fluid-flow-relevant pressure gradients produced by metamorphic reactions within Europa's silicate portion remain largely unknown but, for a conservative hydraulic diffusivity of $10^{-2} \text{ m}^2/\text{s}$ (ref. 36), may take of order 10^4 – 10^6 years to diffuse through the interior.

Should the pore network be closed off from the ocean and pressurised above hydrostatic conditions, the seafloor rocks would be subject to a lower effective pressure and the stresses needed to drive faulting would be decreased. At a depth of 1 km below the ocean floor, the amount of pressurisation over hydrostatic required for normal faulting is 8 MPa. However, it is unclear what mechanism(s) would drive such overpressurisation on Europa in the absence of tectonic activity, major sediment sources, or present-day dehydration reactions at shallow depth—processes that serve to overpressure subseafloor rocks on Earth³⁷.

Apollo seismic measurements on Earth's Moon detected quakes at depths of 700–1200 km linked to tidal cycles^{38,39}. Despite occurring within an expected plastic deformation regime, some of these deep quakes are consistent with slip on faults showing extremely low apparent values of cohesion and friction⁴⁰, together with small differential stresses that are not predicted by our fault strength calculations. Yet most deep lunar quakes are $M_w < 3$ (ref. 41) and do not appear to be kinematically linked with larger, shallower seismic events, even if some of those more powerful quakes are also tidally induced⁴². Although shallow moonquakes show recurrence intervals of months to years⁴², the frequency at which resolvable (thrust) fault strain is accumulated at the lunar surface is of order tens of millions of years⁴³. Faulting is thus an extremely inefficient means of exposing fresh lunar rock to the surface (see Methods). By direct analogy with the Moon, then, small, tidally-driven seismicity deep in Europa's silicate interior⁴⁰ is not incompatible with a tectonically quiescent ocean floor.

Fluid infiltration and rock reactions within the seafloor might also facilitate fracture propagation⁴⁴. Laboratory experiments show that hydration and carbonation reactions increase permeability when rocks are essentially impermeable ($<10^{-18} \text{ m}^2$) and reaction rates greatly exceed fluid flow rates⁴⁵. This finding challenges the prevailing understanding, based on flow-through experiments on peridotite hydration at 150–300 °C that commonly show permeability reductions by pore clogging^{46,47}. All laboratory serpentinisation experiments demonstrating substantial (>50 vol. %) mineral conversions have been conducted at temperatures ≥ 200 °C (refs. 48,49). Whether such

reactions could enhance permeability at Europa's seafloor temperature (i.e., ~0 °C) remains unknown. If rapid reaction rates are needed to enhance permeability, then feedbacks that facilitate fluid flow would necessarily be short lived.

We do not consider stresses arising from non-synchronous rotation¹⁴ given that we focus only on the moon's silicate interior. Further, we discount impact-induced stresses since it is not clear how often (if at all) an impactor that breaches Europa's ice shell⁵⁰ might plausibly retain sufficient energy to have any effect upon the ocean floor.

Brittle–plastic transition depth

With increasing temperature and pressure in rocks, brittle failure gives way to plastic deformation (thus dramatically decreasing permeability). Establishing the depth of the transition zone between fracturing and crystal plasticity within Europa's silicate interior with brittle and plastic flow strength profiles therefore offers information on whether, for example, mantle flow is able to transmit stresses to the surface and drive failure there (see next Section). To estimate the depth and therefore strength of the brittle–plastic transition (BPT) zone, we consider a strain rate of 10^{-20} s^{-1} , below the upper bound for intraplate strain rates on Earth (i.e., $\sim 10^{-17} \text{ s}^{-1}$; ref. 51), and a thermal gradient of 10 K/km (ref. 52) to reflect an elevated heat flow arising from, for instance, substantial tidal dissipation in the silicate mantle. For these calculations, we treat Europa's bulk silicate interior as altered peridotite (i.e., wet olivine; see Methods).

The intersections of the brittle and plastic strength profiles occur at a sub-seafloor depth of about 56 km (Table 2). At that depth, the brittle lithospheric strength for normal and thrust faults peak at ~51 MPa and ~67 MPa, respectively (Fig. 3). The total lithospheric thicknesses, which we define here as where the ductile strength of the lower lithosphere drops below 10 kPa, is 120 km.

For this same strain rate and thermal gradient, the BPT zone depth for Mercury is ~47 km and for Mars is ~65 km, assuming a dry olivine rheology in both cases⁵². These values bracket Europa's silicate interior, and thus the lithospheres of Mercury and Mars can be thought of as its inner Solar System mechanical cognates. Even so, the thermal gradient we use here is likely unrealistically high: a study of melt production in the European mantle featured a gradient of 3.3 K/km, corresponding to a “high radionuclide content” model (ref. 53), in which the satellite's mantle radionuclide abundances correspond to

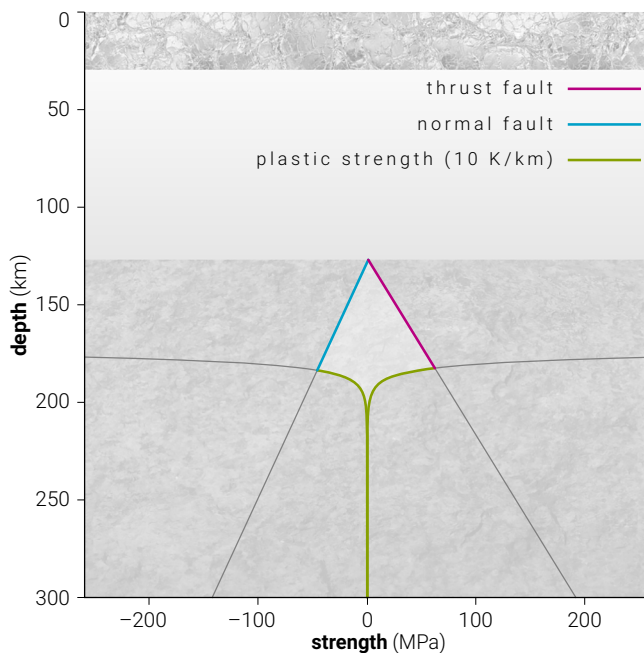


Fig. 3 | Lithospheric strength envelopes for Europa's silicate interior showing the calculated brittle–plastic transition zone depth. The BPT zone depth is given by the intersection of the brittle strength profiles for normal and thrust faults (blue and magenta lines, respectively) and the strength envelope that defines plastic flow in the lower portion of Europa's silicate interior (green lines). The area within the strength envelope is lightly shaded for visualisation purposes. Note that semi-brittle and low-temperature plastic processes are not considered here, which would serve to “round off” the intersections of the brittle and plastic strength curves.

that of LL chondrites⁵⁴. Should Europa's mantle be closer in radiogenic abundances to carbonaceous chondrites⁵⁴, then the interior temperature will be lower, and thus the corresponding BPT zone depth even greater, than the results we report here.

Mantle convective stresses

Stresses in the lithosphere arising from mantle convection, if sufficiently high, are capable of driving slip along existing fractures and even generating new fractures in the brittle lithosphere. If stresses are great enough to exceed the yield strength of the entire lithosphere (as estimated from lithospheric strength profiles such as those we show above), then even plate tectonic-like behaviour could be initiated^{55,56}. We therefore estimate the stresses resulting from mantle convection using scaling relationships developed by ref. 57 (see Methods). For this calculation, we assume that Europa's silicate mantle convects, but under a stagnant-lid regime. This approach provides the strongest test as to whether modern Earth-like plate tectonics could operate, as this phenomenon is thought to be more difficult to initiate from a stagnant-lid state (e.g., ref. 58).

The mantle's interior viscosity sets the vigour of convection and resultant stress scale. Europa's mantle interior viscosity is unknown, and is therefore the largest uncertainty in the scaling law for stress. We explore a range of 10^{18} – 10^{20} Pa s (that used in ref. 53). Increasing mantle viscosity increases the convective stress, but even for a viscosity of 10^{20} Pa s the highest stress we calculate is only about 150 kPa (Fig. 4), consistent with earlier results⁵³—over two orders of magnitude below the minimum lithospheric strength of 51 MPa for normal faults, and less than even the rock strengths we calculate at 100 m depth. The small size of Europa—its mantle volume¹³ is ~57% that of the Moon and 48% that of Mars⁵⁹—means that mantle convective vigour should be low and able to only weakly apply stresses to the lithosphere. It therefore seems unlikely that mantle convection induces brittle

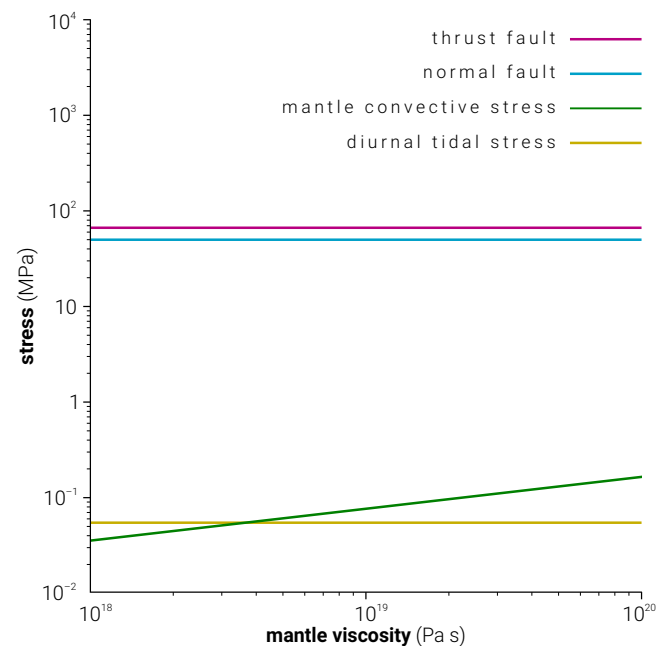


Fig. 4 | Mantle convective stress versus peak lithospheric strength. The differential stress applied to the base of the mechanical lithosphere by mantle convection over two orders of magnitude of mantle viscosity (green line), versus the peak lithospheric strength calculated for thrust and normal faulting (magenta and blue lines, respectively). The stress at the ocean floor from the diurnal tide (~54 kPa) is also shown (gold line).

deformation in Europa's lithosphere at all, much less drives a mobile-lid regime at the seafloor.

Implications of limited seafloor tectonics for European habitability

Our findings suggest that neither tidal stresses, nor global contraction, nor mantle convection, nor serpentinisation is able to drive frequent slip along pre-existing, favourably-oriented faults. These findings hold true even within mechanically fatigued rock, except perhaps at the very uppermost portion of Europa's silicate lithosphere. Since such faulting enables sustained rock–water interactions on Earth²⁵, a lack of tectonic activity implies that any seafloor rocks exposed along fault surfaces will come to attain chemical equilibrium with the overlying ocean. Given that chemotrophic metabolisms exploit chemical disequilibria associated with mixing fluids^{7,60}, a tectonically inert European seafloor does not appear conducive to sustained habitability over geological time.

Moreover, although melt may be generated in Europa's interior⁵³, recent work has found that, because of Europa's size and low melt-generation ability in the present, the maximum buoyancy stress likely to be generated by pooled melt is only ~1–3 MPa (ref. 61)—much lower than the confining strength at the BPT zone depth we calculate (and at the base of the thermal lithosphere calculated by ref. 61). Even if magma overpressures were capable of regularly overcoming the confining strength at such depths, a dyke's ability to propagate upwards is limited to only a few kilometres because of low magmatic buoyancy stresses⁶¹, and thus the prospect of dyke-induced fractures reaching the seafloor seems remote.

The largest hydrothermal flows on Earth's seafloor occur far from sites of volcanic or tectonic activity associated with spreading centres^{62,63}. Instead, massive “ridge-flank” circulation systems discharge reacted fluids at temperatures generally <100 °C and recirculate ocean water in and out of the seafloor every $\sim 10^5$ – 10^6 yr, enhanced by low-permeability sediments, high-permeability aquifers, and

interconnected outcrops^{64–66}. A system of this kind at the European seafloor could facilitate fluid flow³⁴ and promote chemical reactions able to support chemosynthetic ecosystems⁶⁷. Such reactions might even extend beyond ultramafic mineral alteration associated with serpentinisation to include the radiolytic production of H₂ (refs. 68,69), sulphate⁷⁰, and low-molecular-weight carbon compounds⁷¹ driven by U-, Th-, and K-decay. Radiolysis may thus provide a basis for an alien seafloor biosphere even in the absence of substantial fracturing, thermal gradients, or large volumes of unaltered rock because of earlier eras of enhanced geological activity⁷¹—albeit at even lower levels than explored in this study^{68–73}, and so remains to be quantified for Europa.

Whether by serpentinisation or radiolysis, the potential for chemosynthesis-driven habitability requires consideration of other signatures beyond the classic target of O₂ detection⁷⁴. Indeed, the Europa Clipper spacecraft will measure compositional abundances in any plumes that might be present, to test whether water–rock interactions take place today or did in the past⁷⁵. Clipper will also provide unprecedented information on the interior of Europa, including the depth to the rock–water interface, the distribution and amount of tidal heating, and perhaps even whether there is topography on the seafloor^{76,77}. Ultimately, however, the true test of our results here will require directly accessing the ocean and, perhaps one day, the ocean floor itself.

For now, we posit that fracture-moderated habitability at and within Europa's seafloor today is unlikely. Future studies of Europa's modern habitability should focus on the generation of energy for chemoautotrophic life by mechanisms that do not require ongoing seafloor tectonics in the current epoch.

Methods

Lithospheric strength

With recent interior structure models for Europa¹³, we calculate how overburden pressure, P , changes as a function of depth, i.e.,

$$dP/dr = -\rho_r g, \quad (1)$$

where r is body radius (1560.8 km in total), ρ_r is local material density, and g is acceleration due to gravity (1.315 m s⁻² for the outer surface of Europa's ice shell). We use values of 925 kg m⁻³ for ρ_{ice} , 8000 kg m⁻³ for ρ_{metal} (i.e., the moon's metal core), 3429 kg m⁻³ for ρ_{rock} , 2530 kg m⁻³ for ρ_{crust} , and ρ_{water} of 1025 kg m⁻³, after refs. 12,13. We include in our interior structure model a 1-km-thick layer of “crust” at the seafloor that, together with adjusting the value of ρ_{rock} from ref. 13, serves to maintain the mass of the silicate interior. The presence of this layer does not meaningfully alter our findings and is taken to be lithologically and mechanically identical to the underlying silicate mantle, but helps ensure that our results are internally consistent. Gravitational acceleration at a given depth is given by

$$dg/dr = 4\pi G \rho_r - 2(g/r), \quad (2)$$

where G is the gravitational constant⁷⁸.

Our approach assumes that the upper, brittle portion of Europa's silicate interior features pre-existing fractures that, if oriented favourably to stress, will slip. Were such planar weaknesses not present, then the stress required to fracture Europa's silicate portion would be substantially higher than that needed solely for frictional sliding. We calculate the stresses required to enable slip on those pre-existing fractures by the following relations for thrust and normal faults, respectively, using the linearized Mohr–Coulomb failure criterion³⁷:

$$S_H = ((1/\{\mu^2 + 1\})^{0.5} + \mu)^2 \times \sigma^* + P_p + \sigma_c \quad (3)$$

and

$$S_h = ((1/\{\mu^2 + 1\})^{0.5} - \mu)^2 \times (\sigma^* - \sigma_c) + P_p, \quad (4)$$

where S_H and S_h are the maximum and minimum horizontal stresses, μ is the coefficient of static (or internal) friction, P_p is the pore fluid pressure, σ^* is the effective pressure, and σ_c is the unconfined compressive strength of the rock³⁷.

Rather than calculating a specific porosity–depth profile, we regard pore fluid pressure at a given depth to correspond to the effective pressure, σ^* , at that depth, i.e.,

$$\sigma^* = \sigma_1 - P_p, \quad (5)$$

where σ_1 is the maximum principal compressive stress (i.e., S_H for thrust faults and S_h for normal faults). As a result, our model intrinsically assumes that pore space remains throughout the brittle rock volume and that these pores are hydraulically connected to the overlying ocean. Without this assumption, the rock strength would dramatically strengthen as permeability decreases with depth.

Because we assume the effective pressure at the seafloor to be equal to pore fluid pressure, lithospheric shear strength at the rock–water boundary is only that of the rock's unconfined compressive strength, which is related to cohesion, C_0 , by the relation⁷⁹

$$\sigma_c = 2C_0((1/\{\mu^2 + 1\})^{0.5} + \mu), \quad (6)$$

where $C_0 = 1.2$ MPa (after ref. 80). Finally, we employ a value of μ for an altered (0.15, e.g., chrysotile⁸⁰) lithology, representing rock that, having been mineralogically hydrated by interaction with seawater (perhaps during an earlier phase of heating of the entire interior¹⁰), is mechanically weaker and so more likely to undergo faulting.

To validate our lithospheric strength model, we also apply this approach to Enceladus and Earth. For Enceladus, we use an interior structure model based on ref. 13, featuring 50 km of ice and 13 km of water. Calculated rock strengths at a sub-seafloor depth of 1 km require 2.9 MPa for thrust faults and 2.1 MPa for normal faults. Earth is modelled with an ocean depth of 3.68 km, altered seafloor rock, and a 20 K/km temperature gradient. Brittle lithosphere strengths 1 km below the seafloor are determined to be 15 MPa and 11 MPa for thrust and normal faults, respectively.

Stress from the diurnal tide

Tidal extensional stresses develop at two nodes at equatorial latitudes⁸¹, leading to the prospect of extensional structures developing there—optimal for exposing fresh rock to the ocean. The maximum stress from a tidal bulge is¹⁴

$$\sim 3eEH/r, \quad (7)$$

where e is the eccentricity of the satellite's orbit, E is the Young's modulus, in turn given by

$$E = 2S(1 + \nu), \quad (8)$$

in which S is the shear modulus and ν is Poisson's ratio, and where H is the height of the tidal bulge and r is the radius of the satellite. We use values of 70 GPa for S (ref. 82) and 0.28 for the Poisson's ratio⁸³, appropriate for peridotite. The height of the tidal bulge is given by

$$H = h_2 r (M_p/M_s)(r/a_s)^3, \quad (9)$$

where h_2 is the solid body tidal Love number, M_p and M_s are the masses of Jupiter and the silicate interior of Europa, at 1.898×10^{27} kg and 4.43×10^{22} kg, respectively, and a_s is the semi-major axis of the satellite,

with a value of 671,100 km (ref. 14). The value of h_2 for the interior is approximated to first order as

$$h_2 \approx (5/19)(g\rho r_{\text{int}}/S), \quad (10)$$

where g is the gravitational acceleration at the seafloor, ρ is the bulk density of the solid interior, and r_{int} is the radius of the interior⁸⁴.

For Europa, an eccentricity of $e = 0.355$ is required to produce a differential stress of 2.1 MPa (that needed to enable a normal sense of slip along favourably oriented fractures at a depth of 100 m below the seafloor)—an eccentricity much higher than the present value of 0.009.

Stresses from global contraction

The stress state arising from global contraction leads to horizontal shortening of the surface, forming thrust faults if that stress matches the corresponding yield strength of the upper, brittle lithosphere⁴⁹. Although thrusts would not be as efficient in exposing fresh rock to the ocean as joints or normal faults, such structures could still plausibly allow the cycling of seawater through the upper seafloor if critically stressed (cf. ref. 24). The stress from contraction has a value of

$$2S[(1+\nu)/(1-\nu)](\Delta r/r_s), \quad (11)$$

where Δr is the change in radius of the satellite⁸⁵. For a 980-km-thick silicate mantle and a 475-km-thick metallic core for Europa¹³, the required decrease in radius of the solid interior to produce the stresses needed for thrust faults to slip (178 MPa at a depth of 100 m beneath the seafloor) is 1 km (−0.07%). Future thermal evolution models should seek to establish whether such a change in radius is plausible for the European interior, over what time frame global contraction might operate, and if and when the rate of radius change and thrust fault formation is permissive of near-surface fluid flow.

Lunar tectonic resurfacing rates

To first order, we estimate the rate at which shallow quakes could tectonically “resurface” the Moon via the exposure of new crustal material along fault normal planes. Most such quakes are $M_w < 3$ (ref. 41), but for this purpose we presume shallow lunar quakes are regularly $M_w \approx 6$. This quake magnitude is typically associated with a fault rupture length of ~10 km and slip of ~50 cm. Under the assumption that all shallow moonquakes break the surface, a conservative upper bound then of new rock exposed through faulting is $\sim 10^{-3}$ to 10^{-2} km²/yr. Such an exposure rate, even though substantially overestimating lunar fault slip, would require billions of years to resurface a major portion of the Moon given current seismicity, and is qualitatively consistent with today’s largely tectonically quiet lunar surface.

Brittle–plastic transition depth

To establish the depth of the transition zone between fracturing and crystal plasticity within Europa’s silicate interior, we calculate ductile stress by the relation

$$\dot{\epsilon} = A \sigma^n C_{\text{OH}}^r \exp((-E_v + PV)/RT), \quad (12)$$

where $\dot{\epsilon}$ is strain rate, A is a constant, σ is deviatoric stress, n is the stress exponent, C_{OH} is water fugacity, r the water fugacity exponent, E is activation energy, V is the activation volume, P is pressure (which we take to be lithostatic), R is the universal gas constant, and T is temperature⁸⁶. This method of calculating the BPT zone depth is widely used in Earth and planetary science (e.g., refs. 52,87,88).

We use values appropriate to wet olivine deforming by dislocation creep, where $A = 89 \text{ MPa}^{-3.5} \text{ s}^{-1}$, $n = 3.5$, $C_{\text{OH}} = 100$, $r = 1.2$, $E_v = 480 \text{ kJ/mol}$, and $V = 1.1 \times 10^{-5} \text{ m}^3/\text{mol}$ (ref. 89). Temperature gradient and strain rate are 10 K/km (after ref. 52) and 10^{-20} s^{-1} , respectively. (These values are listed in Table 1).

Stresses from mantle convection

Reference 57 developed scaling laws for lithospheric stress resulting from stagnant-lid convection. Two scaling laws were presented in that study, one for a regime with large lid slopes and another for a flat lid. We use the latter model, which results from time-dependent convection as expected for planetary bodies. The scaling law for stress, τ , is

$$\tau = c_1 \rho \alpha g \Delta T_{\text{rh}} \delta_{\text{rh}}, \quad (13)$$

where $c_1 \approx 0.4$ is an empirical constant, ρ is the density of the mantle, α is the thermal expansion coefficient, g is gravity, ΔT_{rh} is the temperature difference across the rheological sublayer (the layer at the base of the stagnant lid that participates in convection), and δ_{rh} is the thickness of that rheological sublayer. Values for density, the thermal expansion coefficient, and gravity are all taken from those provided in ref. 53. The rheological temperature difference ΔT_{rh} and rheological sublayer thickness δ_{rh} are given by⁹⁰

$$\Delta T_{\text{rh}} = (T_i - T_s)/\theta, \quad (14)$$

where T_i is the interior temperature of the mantle, T_s is the surface temperature at the top of the silicate layer, and θ is the Frank–Kamenetskii parameter, which defines the viscosity structure across the layer⁹¹ and is defined as

$$\theta = (E_v(T_i - T_s))/RT_i^2, \quad (15)$$

where, in turn, R is again the universal gas constant, and E_v the activation energy for viscosity. The rheological sublayer thickness is found to follow the scaling relationship⁹⁰

$$\delta_{\text{rh}} = c_2 d (\{Ra_i/\theta\})^{-1/3}, \quad (16)$$

where $c_2 \approx 2$ is an empirical constant, d is thickness of the mantle, and Ra_i is the internal Rayleigh number, itself given by

$$Ra_i = (\rho \alpha g (T_i - T_s) d^3)/(\kappa \mu_i), \quad (17)$$

with α as the thermal expansion coefficient again, κ is thermal diffusivity, and μ_i is the viscosity of the mantle at the interior temperature T_i . Combining the above equations gives a final scaling relationship for stress,

$$\tau = c_1 c_2 (\kappa \mu_i / d_2) (\{Ra_i/\theta\})^{2/3}, \quad (18)$$

which we use to calculate convective stress for Europa. The Frank–Kamenetskii parameter is estimated as $\theta = 18.7$ assuming an activation energy of 300 kJ/mol, a mantle temperature of $T_i = 1600 \text{ K}$, and a (seafloor) surface temperature of $T_s = 273 \text{ K}$ (ref. 53). Other assumed parameters are $\rho = 3500 \text{ kg/m}^3$, $\alpha = 3 \times 10^{-5} \text{ 1/K}$, $g = 1.3 \text{ m/s}^2$, $d = 800 \text{ km}$, and $\kappa = 10^{-6} \text{ m}^2/\text{s}$. Mantle interior viscosity, μ_i , is varied over the range 10^{18} – 10^{20} Pa s as explained in the main text.

Seafloor volcanism

To evaluate the plausibility of volcanism at Europa’s seafloor, we constructed for a separate, related study⁶¹ a model of Europa’s mantle in the geodynamic simulation code StagYY⁹², within which we designed and implemented a new melt-extraction routine. The mantle model is a 2D spherical annulus 821 km in thickness with temperature ranging from 273 to 1600 K. Heat is generated within the model by radioactive decay⁹³ and tidal dissipation⁹⁴. We benchmark internal heating to conditions in the models of ref. 53 to ensure consistency with previous results. Melting occurs along a Herzberg–Bohler solidus curve⁹⁵ and is then transported upwards through the viscous portion of the mantle via Darcy pore-space flow along grain boundaries. When

Table 3 | Values for Earth model parameters

Model parameter	Value
Geometric properties	
Radius (km)	6378
Surface gravitational acceleration (m/s ²)	9.8
Ocean depth (km)	3.682
Radius of silicate mantle (km)	2888.3
Radius of iron core (km)	3486
Mechanical properties	
Density of ocean (kg/m ³)	1030
Density of silicate mantle (kg/m ³)	4500
Density of iron core (kg/m ³)	10,650
Rock cohesion (MPa)	1.2
Rock friction coefficient	0.15
Thermal gradient (K/km)	20
Strain rate (s ⁻¹)	1 × 10 ⁻¹⁵
Dry olivine dislocation flow law (after ref. 66)	
Material parameter, A (MPa ⁻³ s ⁻¹)	110,000
Stress exponent, n	3.5
Activation energy, Q _c (J/mol)	530,000
Hydrogen concentration, C _{OH} (H/10 ⁶ Si)	1
C _{OH} coefficient, r	1
Specific volume, V (m ³ /mol)	14 × 10 ⁻⁶

melt reaches the base of the thermal lithosphere (which we define as 1400 K), it is then evaluated by the new dike–melt extraction treatment based on ref. 96. To erupt, the magma must then satisfy two criteria: 1) magma must exert sufficient buoyancy stress at the base of the lithosphere to match its yield strength and so initiate a dyke; and 2) that dyke must propagate upward through the lithosphere faster than conductive cooling solidifies and halts the intrusion. Full details of this model are described in ref. 61.

Thermal expansion anisotropy

The normal stress, σ_{yy} , on a two-dimensional grain boundary can be derived²⁸ via the following relation:

$$\sigma_{yy}(x) = \frac{E\Delta\alpha(T - T')}{2\pi(1 - \nu^2)} \left(\frac{4L^2}{4L^2 + (2L - x)^2} - \frac{4L^2}{4L^2 + x^2} + \ln \left[\frac{2L - x}{x} \right] - \frac{1}{2} \ln \left[\frac{4L^2 + (2L - x)^2}{4L^2 + x^2} \right] \right), \quad (19)$$

where E is Young’s modulus, α is flaw size, T' is the viscous–elastic transition temperature, L is half grain size, and x is distance⁴⁴. As microfractures accumulate along grain boundaries, cracking occurs when the stress intensity at the tip of the grain exceeds a critical value ($K_I > K_{IC}$) that is in the range 0.2–3.0 MPa m^{1/2}. For instance, the K_{IC} value for olivine is 0.6 ± 0.3 MPa m^{1/2} (ref. 97). The critical stress intensity for a flaw of size α is²⁸

$$K_{IC} = \sqrt{\frac{2}{\pi\alpha}} \int_0^\alpha \frac{\sigma_{yy}(x)\sqrt{x}}{\sqrt{\alpha - x}} dx - P_c \sqrt{\pi\alpha} \quad (20)$$

Model comparison with Earth and Enceladus

As a check of our modelling approach for Europa, we consider two worlds for which the seafloor structure and mechanical properties are either well known (Earth) or have been proposed on the basis of comprehensive spacecraft observations (Saturn’s diminutive icy moon Enceladus). We list the parameters and associated values for each body in Tables 3 and 4. The question of whether fault slip can occur in Earth’s seafloors is a complex one, since laboratory tests suggest that oceanic

Table 4 | Values for Enceladus model parameters

Model parameter	Value
Geometric properties	
Radius (km)	252.3
Surface gravitational acceleration (m/s ²)	0.113
Ice shell thickness (km)	50
Ocean depth (km)	13
Radius of silicate interior (km)	189.3
Orbital properties	
Orbital eccentricity	0.009
Average orbital radius (km)	238,020
Jupiter mass (kg)	5.68 × 10 ²⁶
Mechanical properties	
Density of outer ice shell (kg/m ³)	925
Density of ocean (kg/m ³)	1110
Density of silicate mantle (kg/m ³)	2701
Rock cohesion (MPa)	1.2
Rock friction coefficient	0.15
Thermal gradient (K/km)	10
Strain rate (s ⁻¹)	1 × 10 ⁻²⁰
Wet olivine dislocation flow law (after ref. 96)	
Material parameter, A (MPa ⁻³ s ⁻¹)	89
Stress exponent, n	3.5
Activation energy, Q _c (J/mol)	480,000
Hydrogen concentration, C _{OH} (H/10 ⁶ Si)	100
C _{OH} coefficient, r	1.2
Specific volume, V (m ³ /mol)	11 × 10 ⁻⁶

crust is mechanically stronger than the driving stresses from mantle flow and thus faulting (and, in particular, the initiation of subduction) ought not to occur (e.g., ref. 51). Our calculated results for Earth are given in Table 5. Of note, the absolute differential lithospheric strength we calculate at 100 m depth is 3 MPa in a normal fault regime, and 4 MPa in a thrust fault regime. These values are qualitatively similar to numerical results for subduction initiation of ~3–7 MPa (refs. 57,98). At a depth of 1000 m below the seafloor, these values rise by around a factor of four to 11 MPa and 15 MPa, respectively. Of note, these calculations are for a friction coefficient, μ , of 0.15, corresponding to a basaltic seafloor altered to chrysotile. We revisit the effect of a substantially lower value for μ below.

For Enceladus, given its low mass and surface gravity, its seafloor strength is much lower than that of Europa. The absolute differential lithospheric strength in normal and thrust fault regimes at 100 m depth is 2 MPa and 3 MPa, respectively; the corresponding values at a depth of 1,000 m are functionally identical (Table 6). Even so, Enceladus is unlikely to experience seafloor faulting at all in the present since the maximum tidal stress is only 2 kPa. Although some elastic strain may be expected from Enceladus’ tidal deformation and from compressive stresses arising from secular cooling, no modelled stress is high enough to expose by faulting fresh rock surfaces for reaction with seawater.

In calculating the brittle–plastic transition depth for Earth, we use the values given in Table 3. With a strain rate of 10⁻¹⁵ s⁻¹ (ref. 86) for a dry olivine dislocation rheology, the BPT zone depth is 30 km below the seafloor surface, a value qualitatively consistent with the view that most intraplate earthquakes in oceanic lithosphere occur at temperatures $\lesssim 873$ K, i.e., the 600 °C isotherm⁹⁹. Our model for Enceladus employs the same flow law and component values as for Europa (Table 4), for which we find the Enceladan BPT zone to be at a depth of 65 km beneath the seafloor. This finding, as well as that of relatively low

Table 5 | Model results for Earth

Model result	Value
Lithospheric strength in a normal fault regime (100 m) (MPa)	39
Lithospheric strength in a normal fault regime (1000 m) (MPa)	70
Differential lithospheric strength in a normal fault regime (100 m) (MPa)	3
Differential lithospheric strength in a normal fault regime (1000 m) (MPa)	11
Lithospheric strength in a thrust fault regime (100 m) (MPa)	46
Lithospheric strength in a thrust fault regime (1000 m) (MPa)	96
Differential lithospheric strength in a thrust fault regime (100 m) (MPa)	4
Differential lithospheric strength in a thrust fault regime (1000 m) (MPa)	15
Lithostatic stress (100 m) (MPa)	42
Lithostatic stress (1000 m) (MPa)	81
Effective stress (100 m) (MPa)	3
Effective stress (1000 m) (MPa)	34
Brittle–plastic transition depth in a normal fault regime (km)	36
Brittle–plastic transition depth in a thrust fault regime (km)	36
Peak lithospheric strength for normal faults (at depth below seafloor) (MPa)	322 (36)
Peak lithospheric strength for thrust faults (at depth below seafloor) (MPa)	425 (36)
Mantle stress (10^{18} Pa s) (kPa)	170
Mantle stress (10^{20} Pa s) (kPa)	770
Differential lithospheric strength in a normal fault regime (100 m) (MPa) ($\mu = 3.5 \times 10^{-5}$)	1
Differential lithospheric strength in a thrust fault regime (100 m) (MPa) ($\mu = 3.5 \times 10^{-5}$)	1

Values are rounded to the nearest integer, which serves to give the same depth for the brittle–plastic transition depth for both normal and thrust faults.

Table 6 | Model results for Enceladus

Model result	Value
Lithospheric strength in a normal fault regime (100 m) (MPa)	6
Lithospheric strength in a normal fault regime (1000 m) (MPa)	6
Differential lithospheric strength in a normal fault regime (100 m) (MPa)	2
Differential lithospheric strength in a normal fault regime (1000 m) (MPa)	2
Lithospheric strength in a thrust fault regime (100 m) (MPa)	11
Lithospheric strength in a thrust fault regime (1000 m) (MPa)	11
Differential lithospheric strength in a thrust fault regime (100 m) (MPa)	3
Differential lithospheric strength in a thrust fault regime (1000 m) (MPa)	3
Lithostatic stress (100 m) (MPa)	8
Lithostatic stress (1000 m) (MPa)	8
Effective stress (100 m) (kPa)	23
Effective stress (1000 m) (kPa)	226
Maximum tidal stress (Pa)	2400

Values are rounded to the nearest integer.

fault strengths, is consistent with a substantial portion (~72%) of Enceladus’ silicate interior (i.e., that above the BPT zone) being mechanically unconsolidated, porous, permeable, and hydrated¹⁰⁰.

One solution for Earth to the apparent paradox of a lithosphere substantially stronger than mantle convective stresses is to use much (much) lower friction coefficients^{57,87,101}, possibly from fault weakening by seawater. For our calculated convective stress for Earth (Table 5), the seafloor rocks require a value of μ of 3.5×10^{-5} to undergo failure at a depth of 100 m below the ocean. This is functionally, within our models, the mechanical threshold needed for plate tectonics on Earth. Importantly, even with this friction coefficient applied to Europa, we find that the lithosphere is about 1.3 times stronger than the mantle convective stress (for any of the viscosities we consider). This result suggests that, even with a dramatically lower coefficient of friction,

faults at Europa’s seafloor are probably too strong to undergo deformation from mantle convection sufficient to initiate subduction, such that the silicate lithosphere is in a stagnant lid regime.

For the European seafloor to be in a mobile-lid regime today, our modelling indicates that the rocks there must proportionally weaker than seafloor rocks on Earth. Additional interior heating (e.g., from tidal dissipation) will not account for this weakening, since such heating serves to lower mantle viscosity (Fig. 4). It is not clear to us what geological process(es) might have weakened Europa’s seafloor enough to allow for mantle-driven faulting there at the present.

Data availability

The interior structure model we use for Europa is from Vance et al. (2017), our ref. 13. All other data sources are given in the main text and Methods.

Code availability

The Python script developed for this study is provided in the Supplementary Material as Supplementary Code 1.

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Author contributions

P.K.B. and C.K. conceived of the study. P.V.R. and H.D. developed the code and led the fault-strength and BPT-depth calculations. K.T.C. and C.K. led the rock mechanics elements. S.D.V. and M.M.D. provided the interior structure models and led the thermal expansion anisotropy analysis. J.G.C., D.J.H., N.R.-F., and D.A.W. contributed expertise relevant to seafloor environments and pore fluid pressure conditions. P.S. provided insight into micro-deformation mechanisms and the BPT analysis. M.P.P. led the lunar analogue element of the study. B.J.F. led the mantle convection analysis. A.P.G. and C.M.E. led the seafloor volcanism discussion in the paper. C.R.G. and B.S.L. contributed expertise to the discussion regarding bioavailable nutrients, radiolysis, and future spacecraft observations. P.K.B. led the writing of the manuscript, to which all co-authors contributed, and drafted the figures.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Paul K. Byrne.

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